

Work Portfolio

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Master Thesis

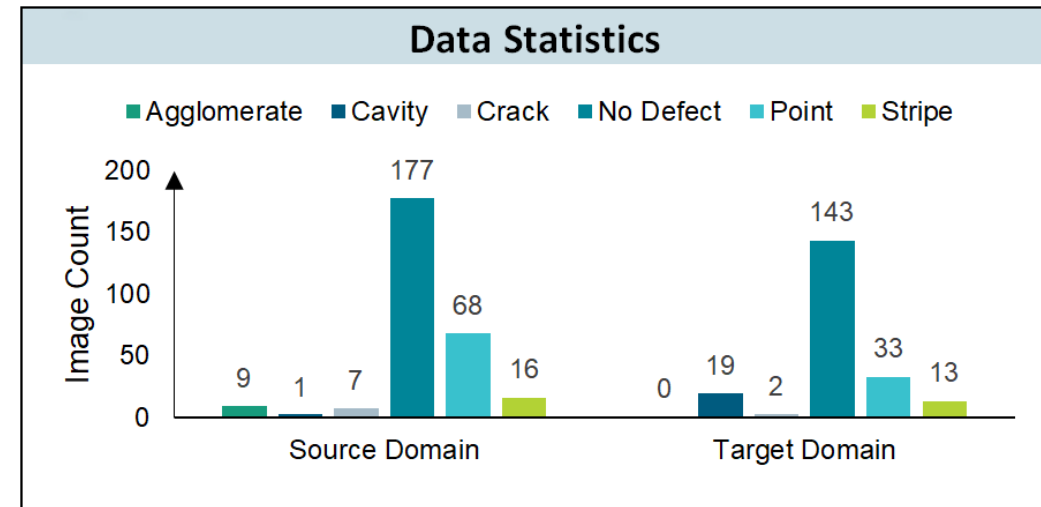
Goal: To train and test *Unsupervised Domain Adaptation Model* for Li-ion Battery Dataset

Tasks

- Literature Review (*Web of Science*)
- Setup & Configuration Management
- Data Pre-processing
- Training Models (*Pytorch*)
- Tuning experiments
- Practical Viability analysis

Challenges

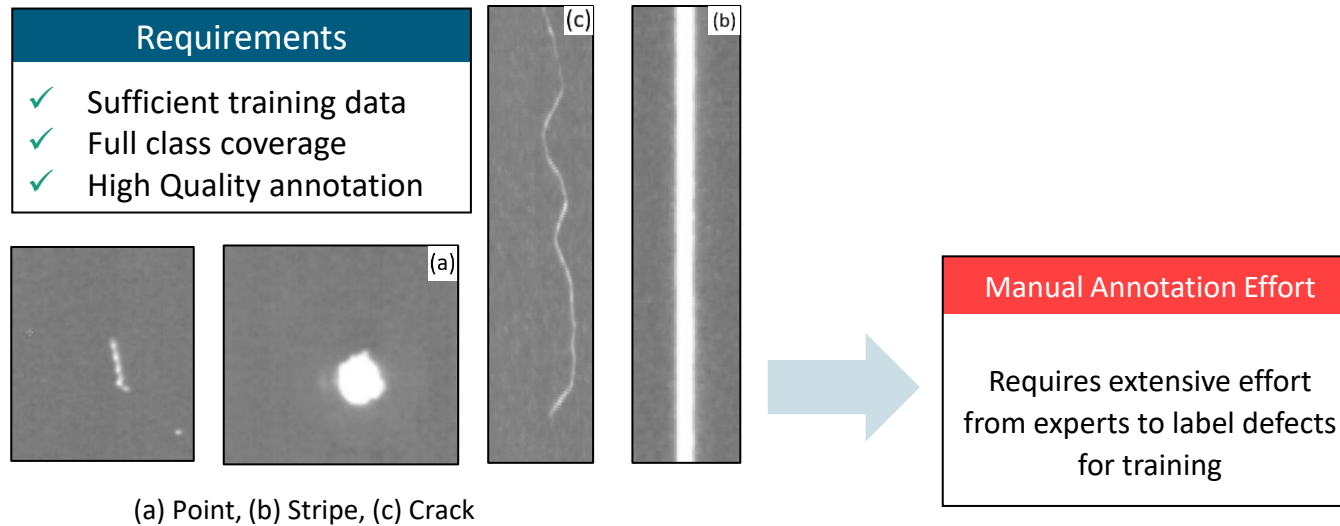
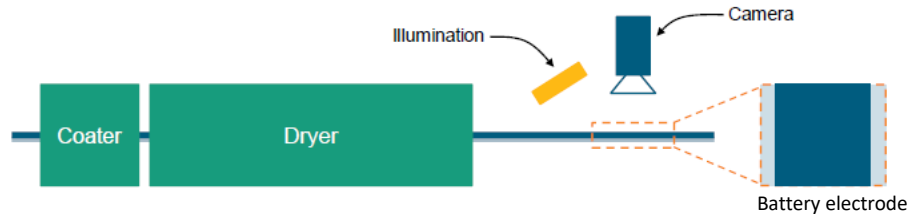
- Highly Unbalanced and very small dataset
- Model Selection (*12 selected*)
- Lot of Experiments



Motivation

The Challenge of Defect Detection in Li-ion battery electrodes

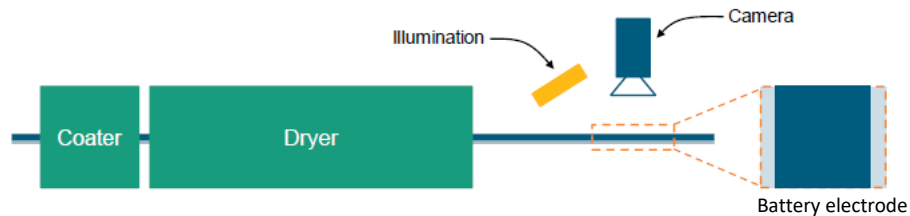
Defect Detection System



Motivation

The Challenge of Defect Detection in Li-ion battery electrodes

Defect Detection System

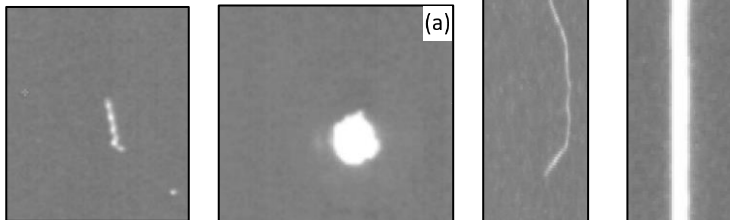


Variations in Production line

- Lighting conditions
- Material composition
- Mechanical re-alignments
- Changed production parameters
- Changed camera configuration

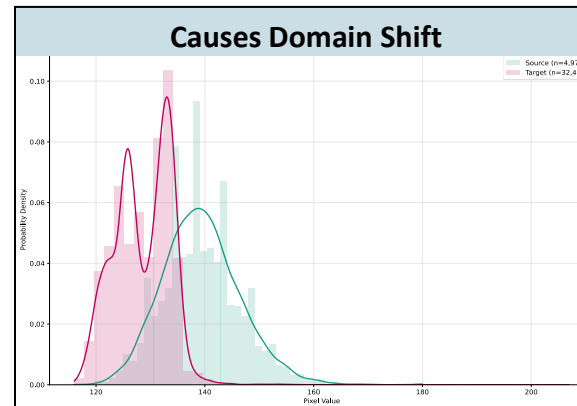
Requirements

- ✓ Sufficient training data
- ✓ Full class coverage
- ✓ High Quality annotation



(a) Point, (b) Stripe, (c) Crack

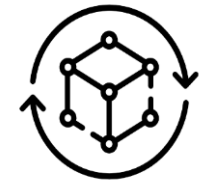
Causes Domain Shift



Performance Drop

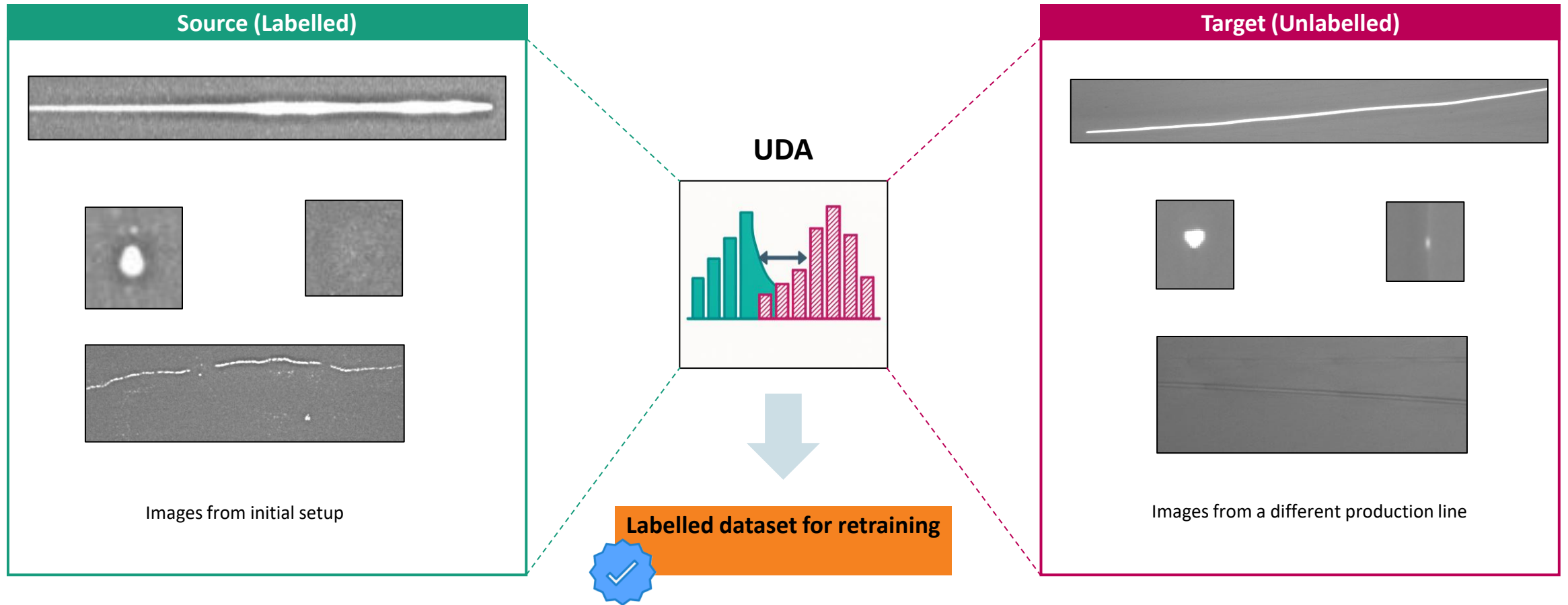


Retrain the model with additional data



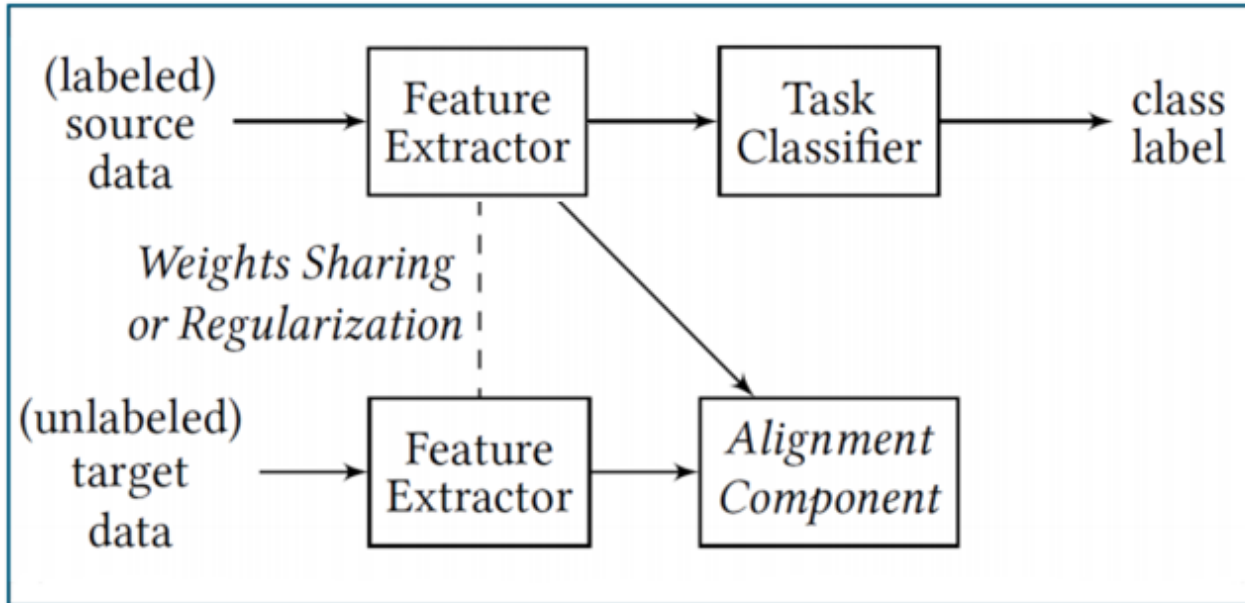
Key Idea

Reducing labelling efforts using UDA



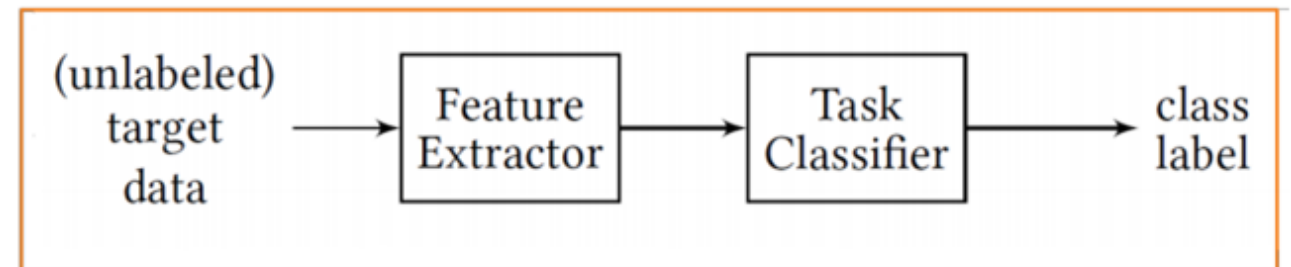
UDA Training Regime & Usage

Learning common feature extractor



← **Step 1:
Training**

**Step 2:
Testing**



Tuning experiments

Full list of planned tuning experiments

		Data Centric				Model Centric						
Options	Models	class balancing	aspect ratio	cropping	augmentation	backbone	lr	lr scheduler	transfer loss	batch size	epoch	base optimizer
	type	discrete	discrete		discrete	discrete	continuous	discrete	continuous	discrete	discrete	discrete
	models	default	default	default	default	ResNet 18	0.01	lamda	0.5	32	50	SGD
	MIC	Undersampling	1:1	center	set 1	ResNet 34	0.05	cos	1	64	20	Adam
	CDTrans	Inverse-weighted		random	set 2	ResNet 50	0.1			128		
	TVT			None	set 1 + set 2	ResNet 101						
Best Model												
1	best	default										
2	best	Undersampling										
3	best	inverse-weighted	default									
4	best	best	[1,, 1,]									
5	best	best	best	center								
6	best	best	best	random								
7	best	best	best	None	default							
8	best	best	best	best	set 1							
9	best	best	best	best	set 2							
10	best	best	best	best	set 1 + set 2							
11	best	best	best	best	best	ResNet34						
14	best	best	best	best	best	ResNet50						
15	best	best	best	best	best	best	0.05					
16	best	best	best	best	best	best	0.1					
17	best	best	best	best	best	best	best	cos				
18	best	best	best	best	best	best	best	best	0,5			
19	best	best	best	best	best	best	best	best	best	64		
20	best	best	best	best	best	best	best	best	best	128		
21	best	best	best	best	best	best	best	best	best	best	20	
22	best	best	best	best	best	best	best	best	best	best	75	
23	best	best	best	best	best	best	best	best	best	best	best	Adam

HiWi Tasks

*At various institutes affiliated with
RWTH Aachen*

Synthetic Data

Goal:

To generate artificial image from Simulation Data

Tasks:

- Data preprocessing
- Image generation
- Image processing
- Visualization

Challenges:

- Porting previous code from MATLAB to python
- Resolving the artifact issue

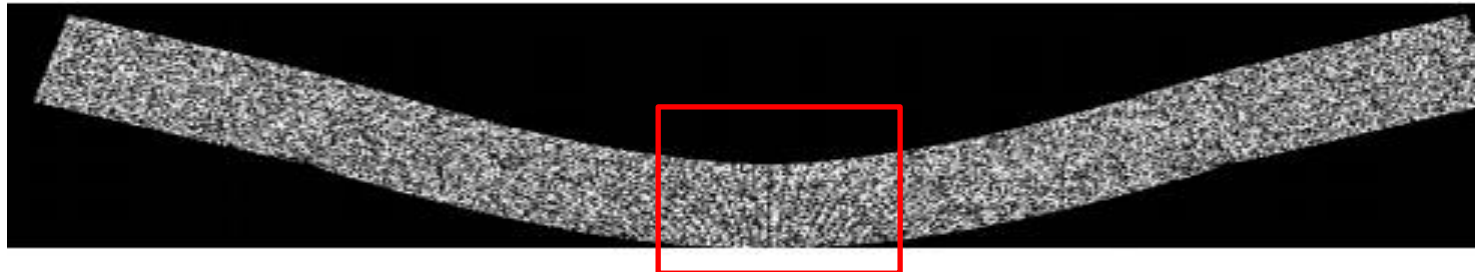
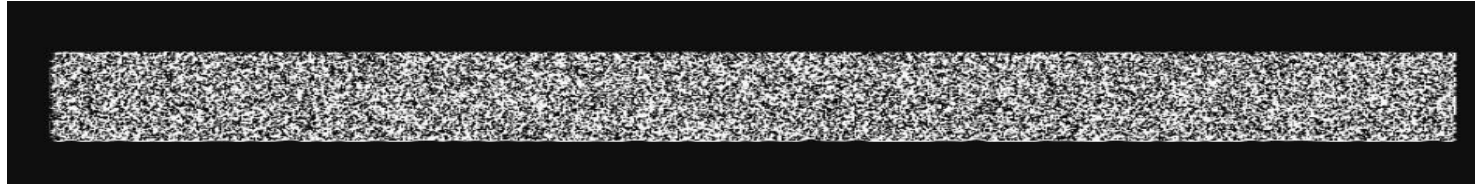
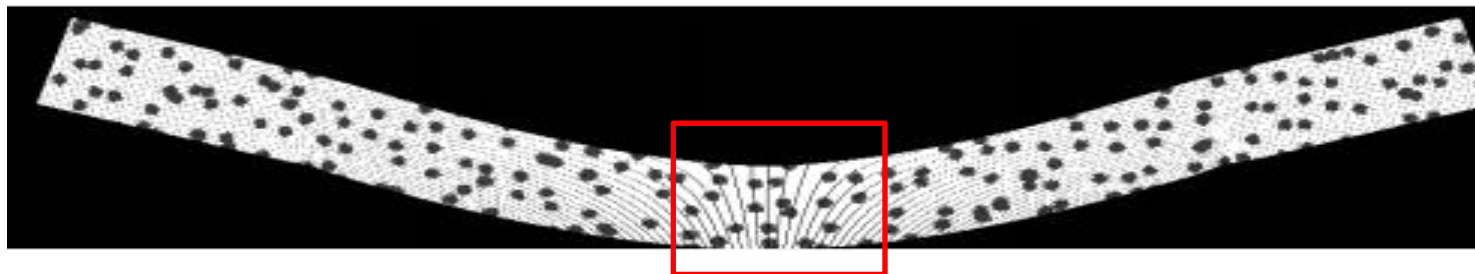
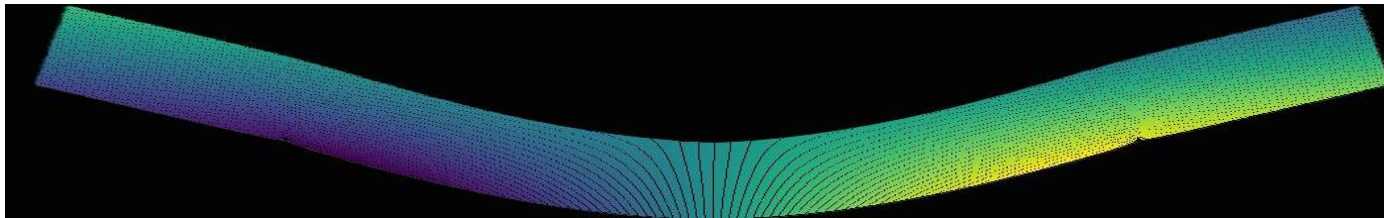
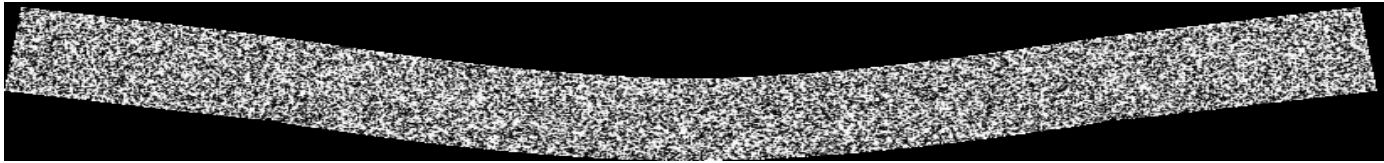
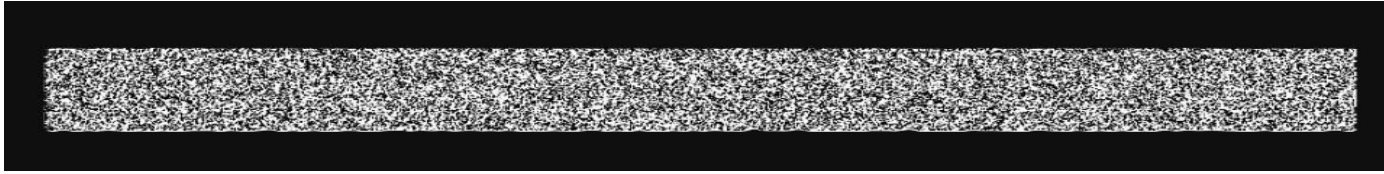


Figure 1 Deformed Speckle Image with Artifacts (example 1)

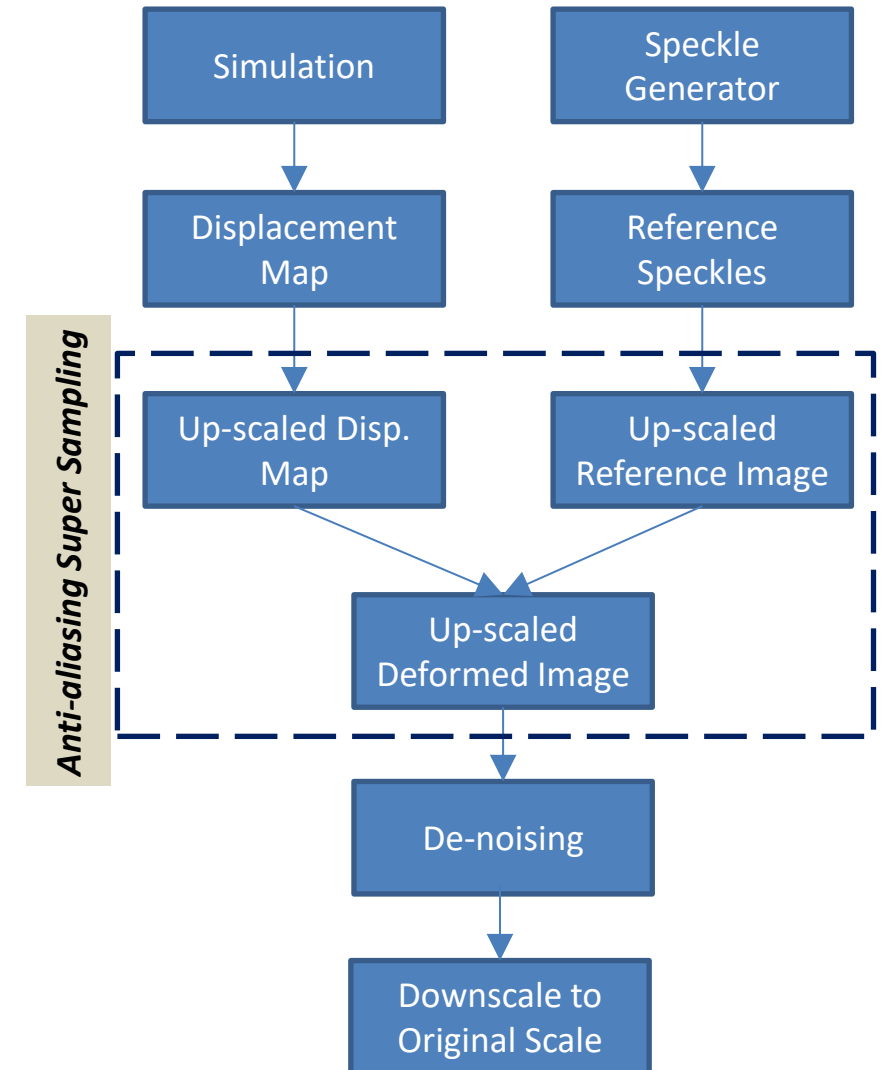


Synthetic Data



Artifact-free image

*RGB image for representation



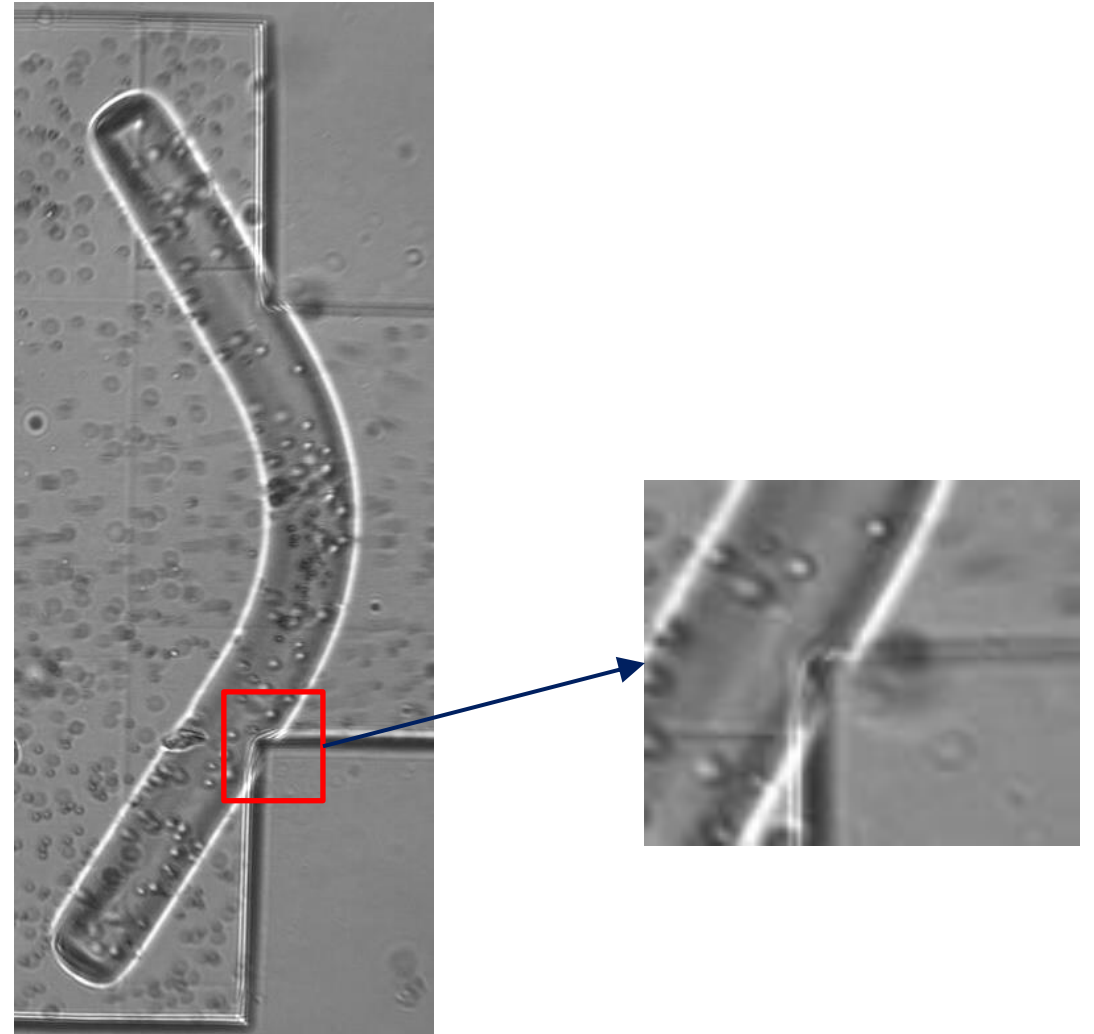
Coding Tasks I

Goal

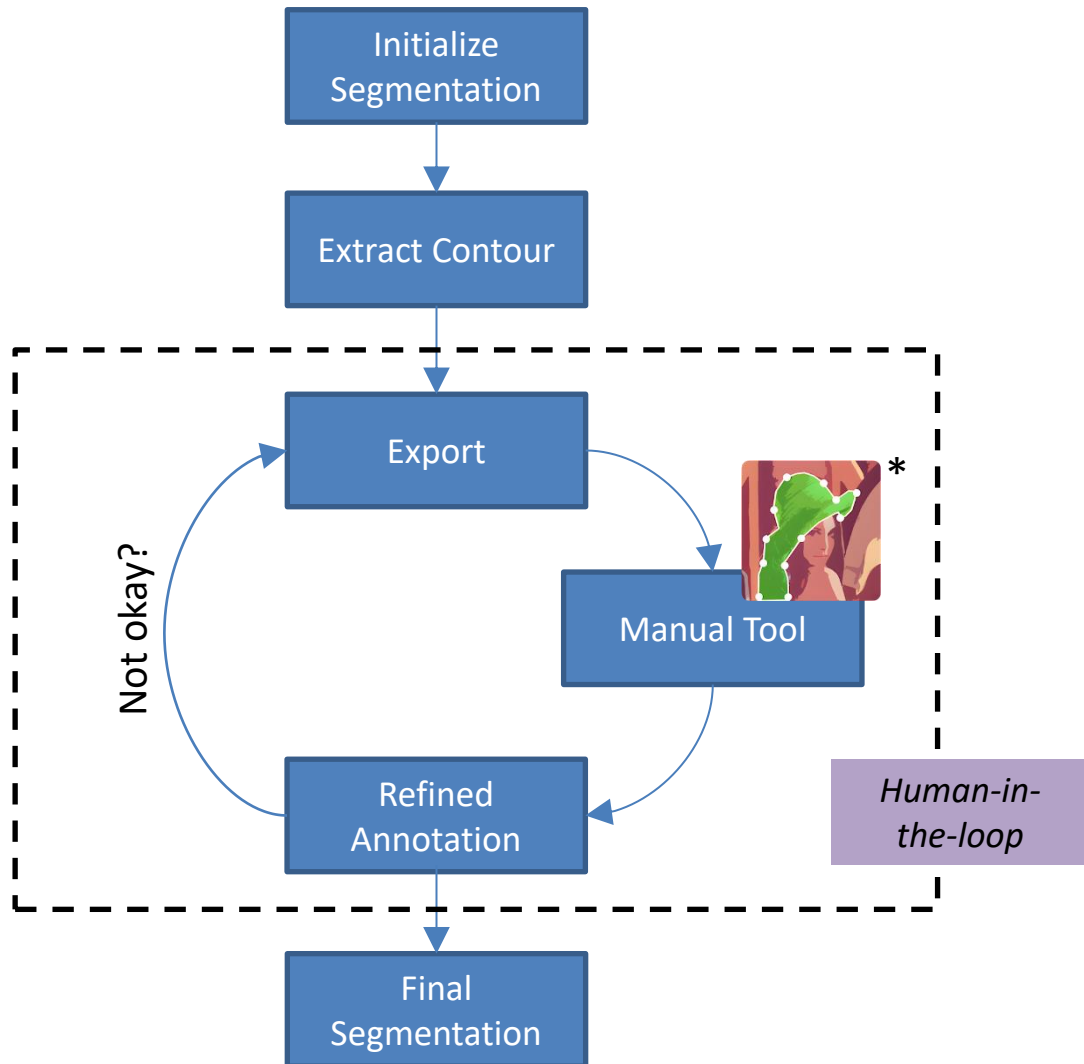
To generate accurate segmentation

Challenge

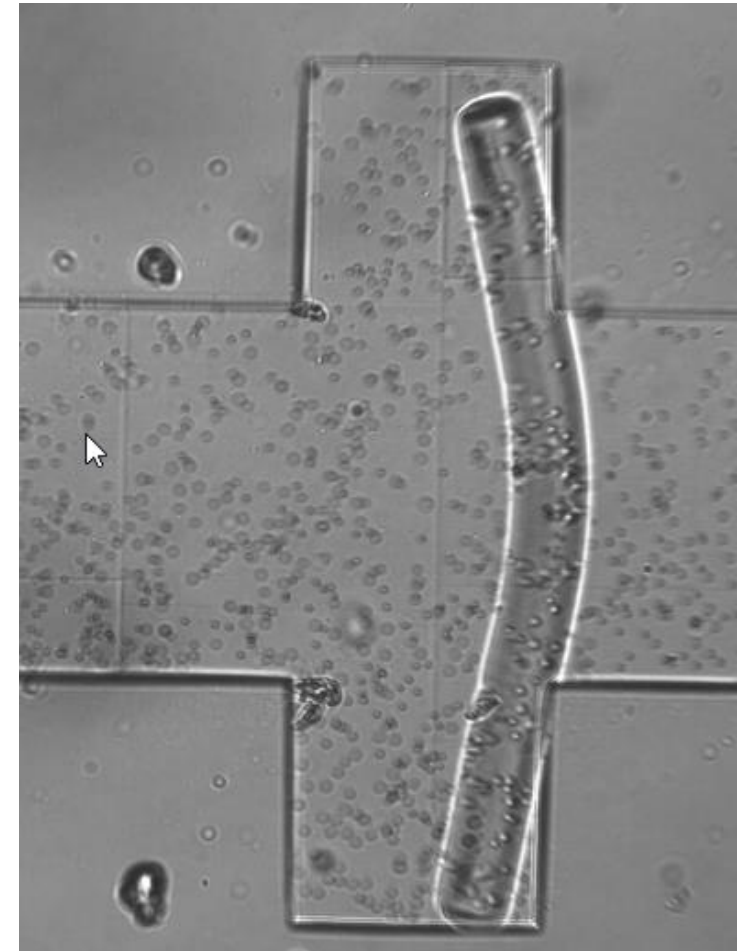
Poor contrast at some regions



Coding Tasks I



**Tool used: Label Me*

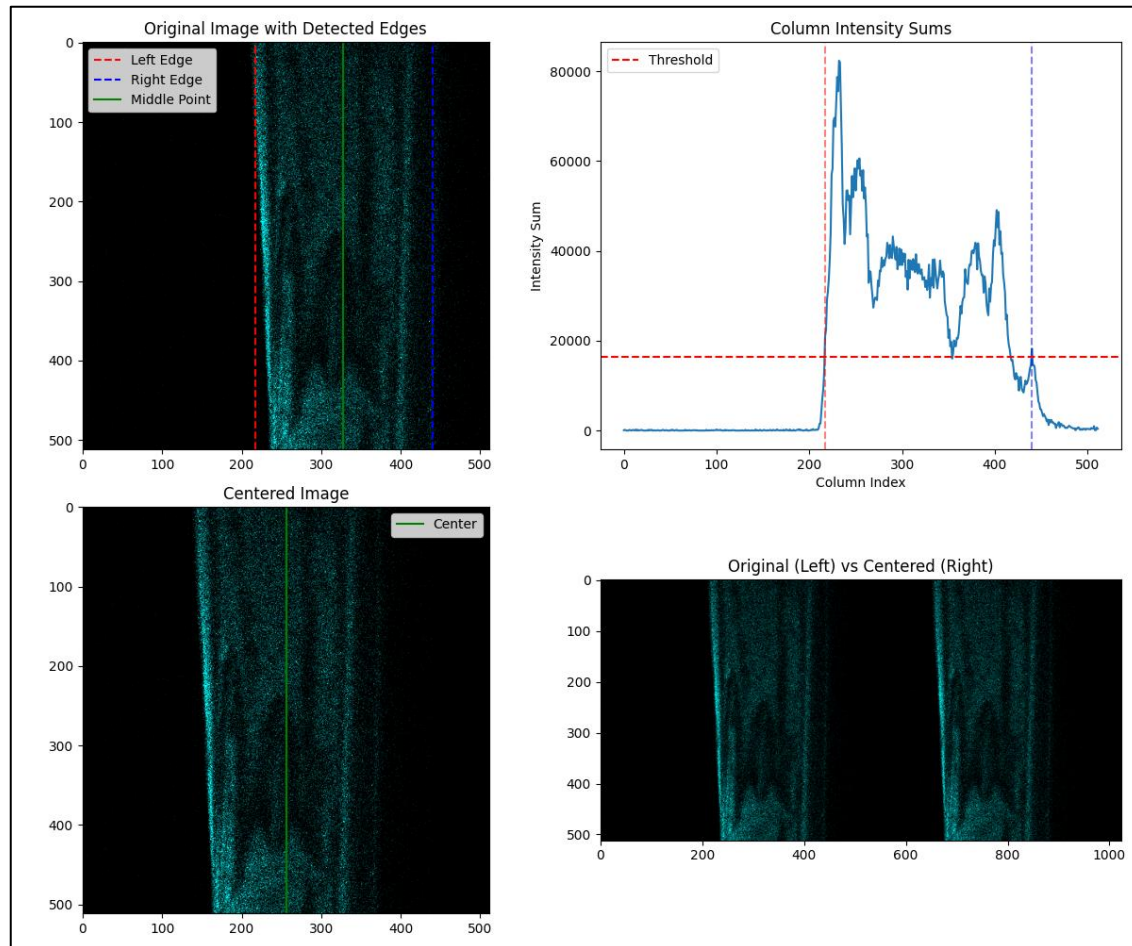


GIF showing process

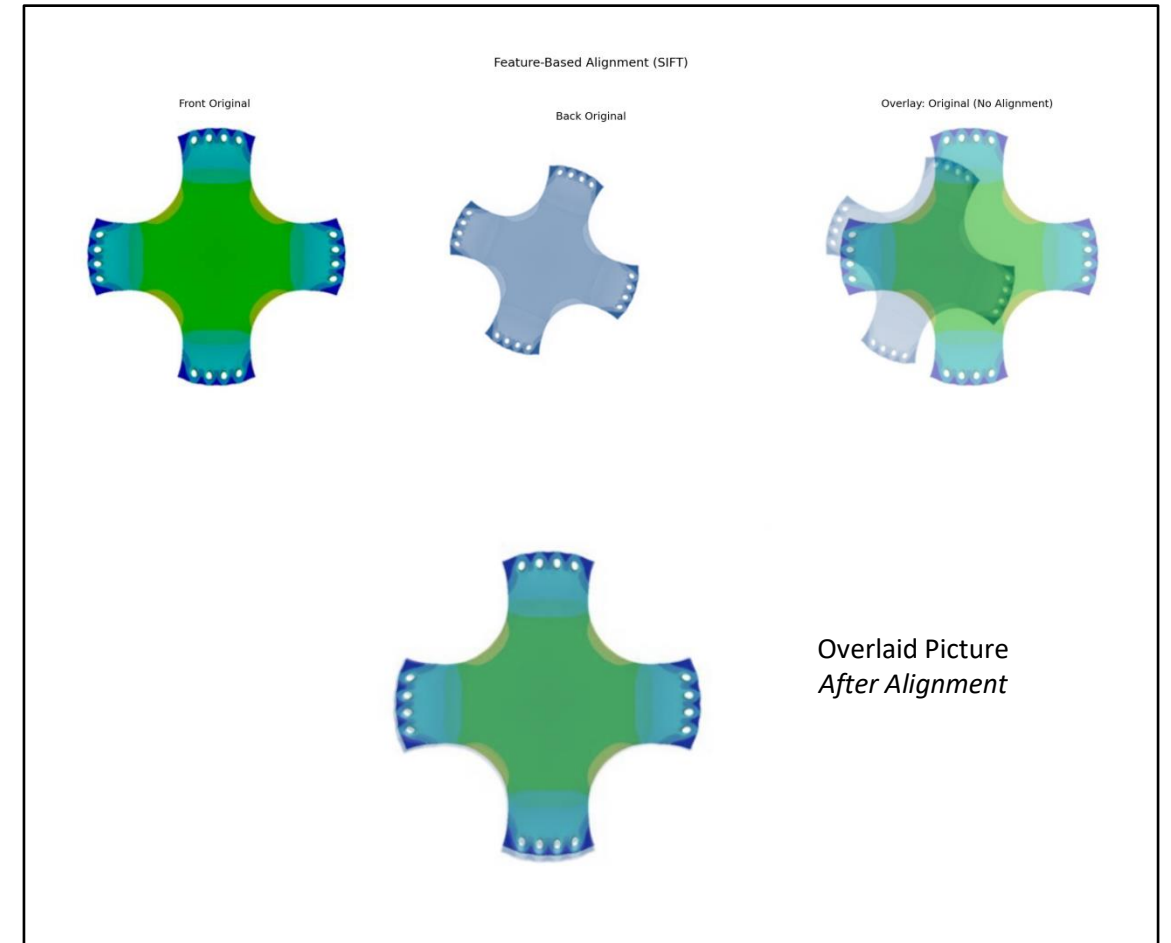


Coding Tasks II

Image Pre-processing for image without any clear edge



Aligning front and back side using feature matching



Coding Tasks III

Tasks

- Automation using python scripts
- Rewriting code for parallel computation (Python)
- Python to Julia
- Maple to Julia
- A small plugin for simulation software in C++

Challenges

Learning new languages

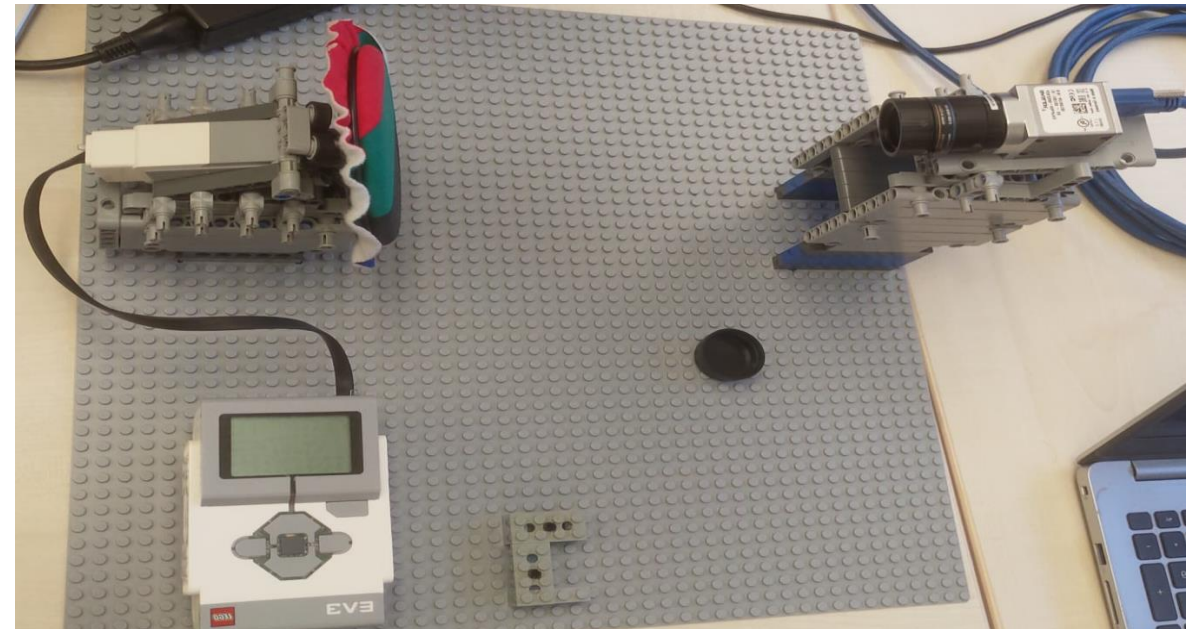
```
start_time = time.time()

args_list = [(i, fintu, dI12, dI12m, I12max, I12maxm, u, uv, phi, phiv, xi, gp) for i in range(4)]

instances_fintu = min(4, mp.cpu_count())

with mp.Pool(processes=instances_fintu) as pool:
    results = pool.map(compute_FintuV, args_list)
```

Pilling Test Prototype



UI using Streamlit

Modified and refactored an user interface to visualize results from the Prototype

Deploy



← Back

 VIEW IMAGES

▼

Available Samples: 0

Sample Number ⓘ
00000

Assessment Stage ⓘ
0

Trial Number ⓘ
1

Loading Weight ⓘ
0

✓ Input parameters valid

 Capture Images and Calculate the Grades

 Results

Export The Results

Reset The Grade

Status

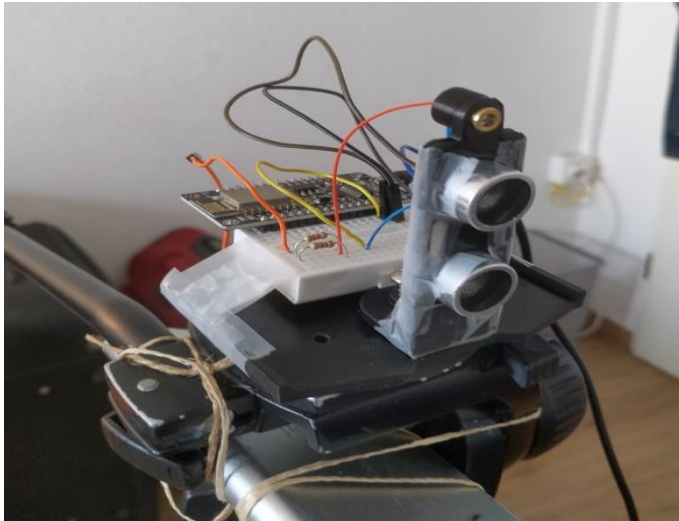
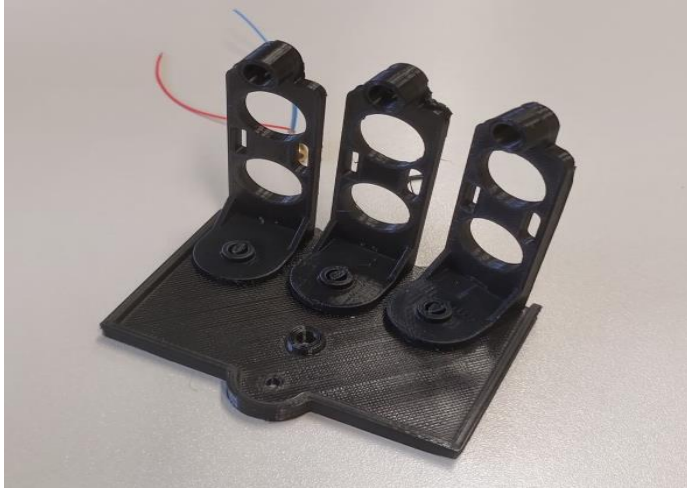
Sample Number 00000, Assessment Stage 0, Trial Number 1

Available Samples
0/8

Results
0/8

 VIEW IMAGES

Ultrasonic Sensor Evaluation



← HC-SR04 with NodeMCU v3



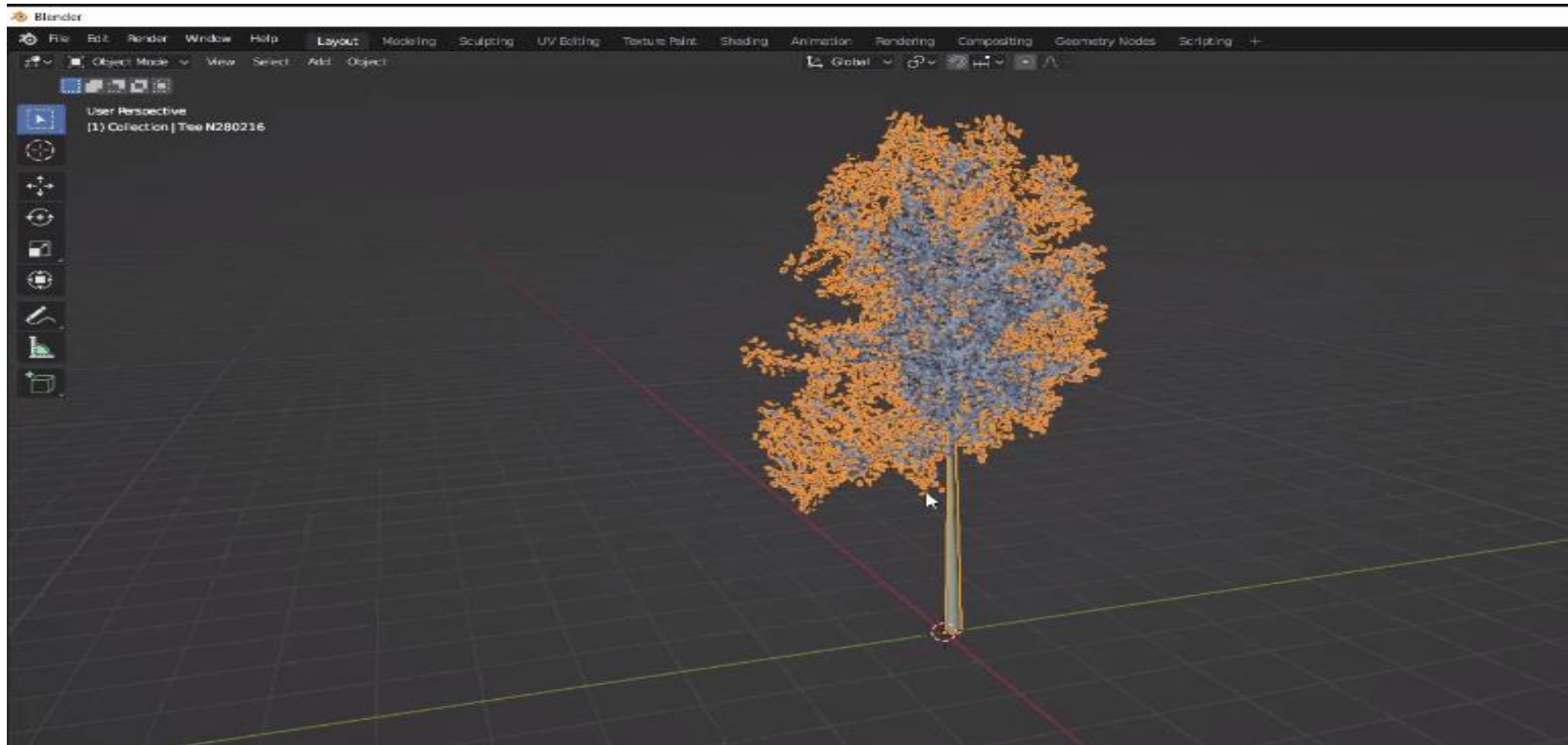
Goal

Evaluate Ultrasonic Sensor for Tree detection

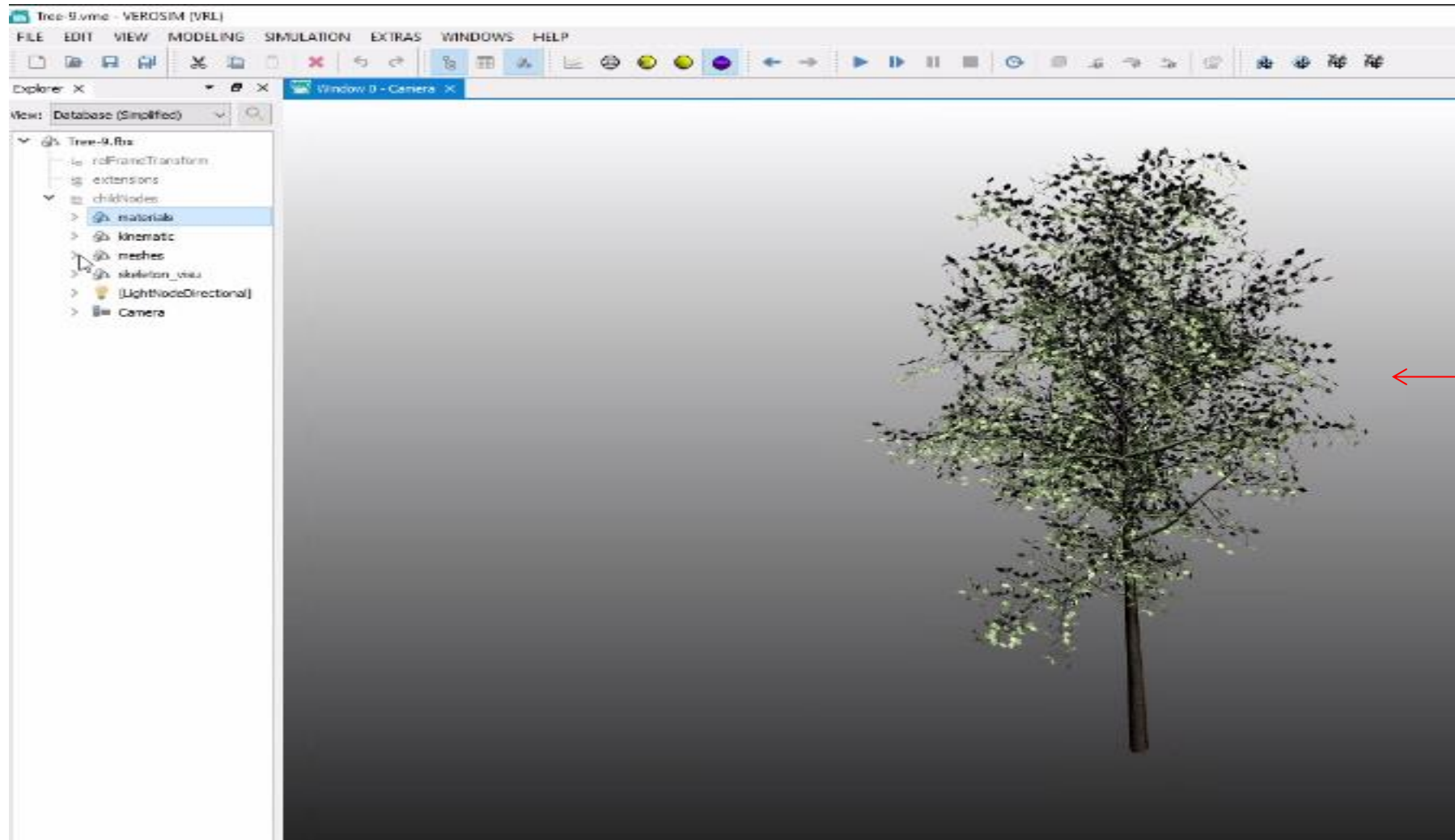
Tasks

- 3D printed Mounting
- Arduino Code
- MQTT setup
- Data Collection

Tree Model Library (Blender to Sim Software)

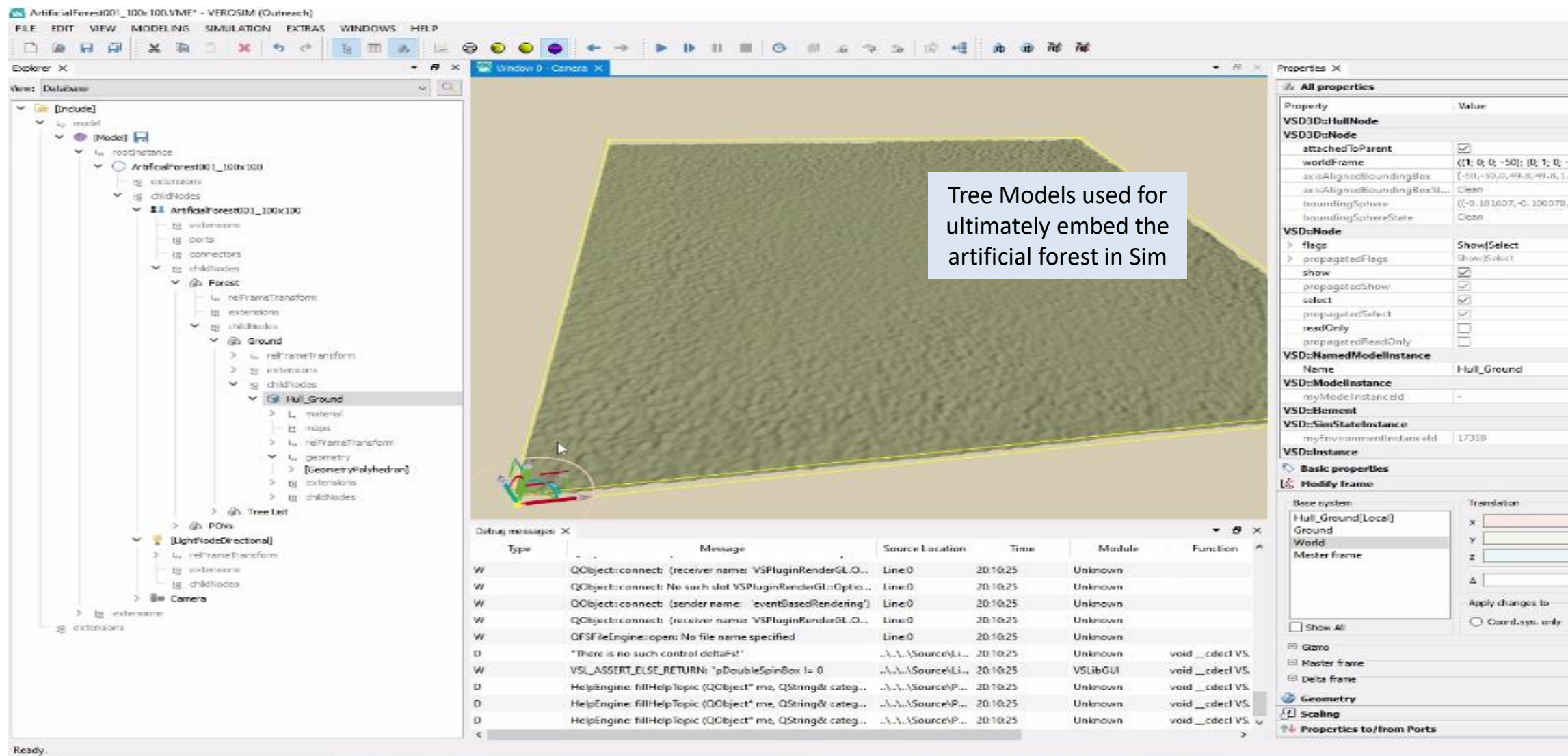


Tree Model Library (Blender to Sim Software)



Ready-to-use Model
for simulation (with
Texture)

Tree Model Library (Blender to Sim Software)



Thank you!

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